Question	Q1	Q2	Q3	Q4	Q5	Q6	Total
Points							

INSTRUCTIONS:

1. The exam will start at **11:00** and will finish at **12:30**. The total time of the exam is **90** minutes.
2. This exam consists of six questions. Please explain your answers step by step.
3. Take **four** digits after the comma/point.
4. Use **radian** when calculating with **trigonometric** functions.

Good luck!

QUESTIONS

1. Consider the following set of three linear equations:

$$\begin{aligned} 2x_1 - 6x_2 + 6x_3 &= 2 \\ 3x_1 - 7x_2 + 13x_3 &= -13 \\ -2x_1 + 2x_2 - 11x_3 &= 21 \end{aligned}$$

   a. Choose which method you will use to solve the problem and create the iterative system in detail. (Either Jacobi iterative method or Gauss-Seidel iterative method.) (7.5 pts)

   b. Carry out the first three iterations of the solution of this system using the method you chose. (7.5 pts)

2. What are interpolation and extrapolation in numerical methods and why do we use them? What is the difference between them? Explain briefly. (5 pts)

x=7

y=54 -36 -12

3. The set of the following five data points is given:

x	1	2	4	5	7
y	54	7	-3	-38	12

a. Determine the fourth-order interpolation polynomial in Newton's form that passes through these points. Calculate the coefficients by using a divided difference table. (15 pts)

b. Use the polynomial obtained in part (a) to determine the interpolated value for **x=3**. (5 pts)

4. The following data shows estimates of the population of a city in selected years between 1970 and 2020:

Year	1970	1980	1990	2000	2010	2020
Population (millions)	1.1	1.4	1.9	2.1	2.8	4

a. Calculate the rate of growth of the population in millions per year for 2020 using the backward difference formula. (10 pts)

b. Using the slope in 2020 from part (a), apply the central difference formula to extrapolate and predict the population in the year **2030**. (10 pts)

5. Let $f(x) = xe^x$.

a. Calculate approximations to $f'(2.3)$ using the backward, forward, and central difference formulas with $h=0.1$, and $h=0.01$. (15 pts)

b. Compare the results with the exact (analytical) derivative (using absolute errors). Determine the best approximation. (5 pts)

6. Consider the function $f(x) = \cos^2(x)$. (Hint: $\cos^2(x) = (\frac{1+\cos(2x)}{2})$)

a. Approximate the integral $\int_0^{\pi} f(x)dx$ using the composite trapezoidal method for $n=4$. (15 pts)

b. Calculate the absolute value error. (5 pts)